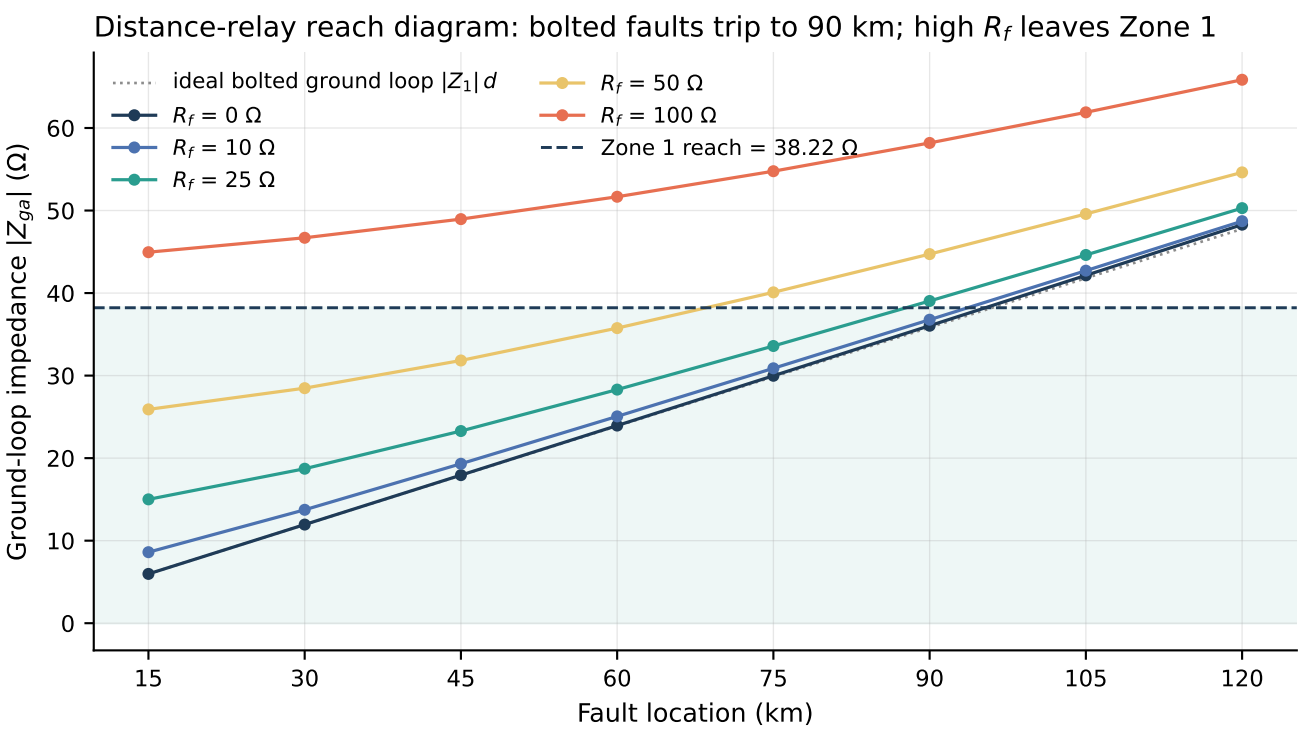
Distance-relay reach diagram: bolted faults trip to 90 km; high R_f leaves Zone 1
ideal bolted ground loop $|Z_1|\,d$
R_f = 0 Ω
R_f = 10 Ω
R_f = 25 Ω
R_f = 50 Ω
R_f = 100 Ω
Zone 1 reach = 38.22 Ω
Ground-loop impedance $|Z_{gal}|$ (Ω)
Fault location (km)
0
10
20
30
40
50
60
15
30
45
60
75
90
105
120